Solution of Tutorial 11

Q.1 (a) $f(x,y) = (x^2+xy)^3$ at $(1,0)$

Solu^n)

$$f_x(x,y) = 3(x^2+xy)^2(2x+y)$$

$$f_x(1,0) = 3(1+0)^2(2\times1+0)$$

$$= 3\times1\times2 = 6$$

$$f_y(x,y) = 3(x^2+xy)^2(x)$$

$$f_y(1,0) = 3(1+0)^2\cdot1 = 3$$

By using limit

$$f_x(1,0) = \lim_{h\to0}\frac{f(1+h,0)-f(1,0)}{h}$$

$$= \lim_{h\to0}\frac{[(1+h)^2+(1+h)0]^3-[1+0]^3}{h}$$

$$= \lim_{h\to0}\frac{(1+h)^6-1}{h} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{h\to0}\frac{6(1+h)^5\cdot1}{1} = 6$$

$$f_y(1,0) = \lim_{k\to0}\frac{f(1,k)-f(1,0)}{k}$$

$$= \cancel{\lim_{k\to0}}\ \cancel{[(1)^2+1\cdot k)^3+1\cdot k]}$$

$$= \lim_{k\to0}\frac{[(1^2+1\cdot k)^3-(1+0)^3]}{k}$$

$$= \lim_{k\to0}\frac{[(1+k)^3-1]}{k} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{k\to0}\frac{3(1+k)^2}{1} = 3$$

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(5) $g(x,y) = \dfrac{x^2+y^2}{xy}$ at $(\sqrt{2},\sqrt{2})$ (2)

$$g_x(x,y) = \frac{(xy)\frac{\partial}{\partial x}(x^2+y^2) - (x^2+y^2)\frac{\partial}{\partial x}(xy)}{(xy)^2}$$

$$= \frac{xy \cdot 2x - (x^2+y^2)y}{(xy)^2}$$

$$g_x(\sqrt{2},\sqrt{2}) = \frac{2\cdot 2\cdot\sqrt{2} - (2+2)\sqrt{2}}{(\sqrt{2}\sqrt{2})^2} = 0$$

$$g_y(xy) = \frac{(xy)\frac{\partial}{\partial y}(x^2+y^2) - (x^2+y^2)\frac{\partial}{\partial y}(xy)}{(xy)^2}$$

$$= \frac{xy \cdot 2y - (x^2+y^2)x}{(xy)^2}$$

$$g_y(\sqrt{2},\sqrt{2}) = \frac{2\sqrt{2}(\sqrt{2})^2 - (2+2)\sqrt{2}}{(\sqrt{2}\sqrt{2})^2} = 0$$

(2) $$f(x,y) = \begin{cases} \dfrac{xy}{\sqrt{x^2+y^2}} & x^2+y^2 \neq 0 \\ 0 & x=y=0 \end{cases}$$

$$f_x(0,0) = \lim_{h\to 0} \frac{f(0+h,0) - f(0,0)}{h}$$

$$= \lim_{h\to 0} \frac{0-0}{h} = 0$$

$$f_y(0,0) = \lim_{h\to 0} \frac{f(0,0+k) - f(0,0)}{k} = \lim_{k\to 0} \frac{0-0}{k} = 0$$

$$f(0+h,0+k) - f(0,0) = h f_x(0,0) + k f_y(0,0) + h\epsilon_1 + k\epsilon_2$$

$$\frac{hk}{\sqrt{h^2+k^2}} - 0 = h\cdot 0 + k\cdot 0 + h\epsilon_1 + k\epsilon_2$$

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$$\Rightarrow \frac{hk}{\sqrt{h^2+k^2}} = h\epsilon_1 + k\epsilon_2$$

Let $h = k$

$$\Rightarrow \frac{h^2}{\sqrt{2h^2}} = h(\epsilon_1+\epsilon_2)$$

$\Rightarrow \epsilon_1+\epsilon_2 \not\to 0$ as $h \to 0$

So f is not diff. at $(0,0)$

(b) $$g(x,y) = \begin{cases} \dfrac{2x^3+3y^3}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

$$g_x(0,0) = \lim_{h\to 0} \frac{g(h,0)-g(0,0)}{h}$$

$$= \lim_{h\to 0} \frac{\frac{2h^3}{h^2} - 0}{h} = 2$$

$$g_y(0,0) = \lim_{k\to 0} \frac{g(0,k)-g(0,0)}{k}$$

$$= \lim_{k\to 0} \frac{\frac{3k^3}{k^2} - 0}{k} = 3$$

$$g(h,k) - g(0,0) = h g_x(0,0) + k g_y(0,0) + h\epsilon_1 + k\epsilon_2$$

$$\frac{2h^3+3k^3}{h^2+k^2} - 0 = 2h + 3k + h\epsilon_1 + k\epsilon_2$$

Let $k = h$

$$\frac{5h^3}{2h^2} = 5h + h(\epsilon_1+\epsilon_2)$$

$$\frac{5h}{2} = 5h + h(\epsilon_1+\epsilon_2)$$

$$\frac{5}{2} = 5 + \epsilon_1+\epsilon_2 \Rightarrow \epsilon_1+\epsilon_2 = -5/2$$

$\epsilon_1+\epsilon_2 \not\to 0$ as $h \to 0$

So not diff at $(0,0)$

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3 (a) $f(x,y) = x^3 + y^3 + 3x^2 - 3y^2 - 8$

$f_x = 3x^2 + 6x \qquad f_{xy} = 0$

$f_{xx} = 6x + 6$

$f_y = 3y^2 - 6y$

$f_{yy} = 6y - 6$

for critical points put $f_x = 0 \Rightarrow 3x(x+2) = 0 \Rightarrow x = 0, x = -2$

$f_y = 0 \Rightarrow 3y(y-2) = 0 \Rightarrow y = 0, y = 2$

Critical points are : $(0,0)\ (0,2)\ (-2,0)\ (-2,2)$

$$D(x,y) = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = \begin{vmatrix} 6x+6 & 0 \\ 0 & 6y-6 \end{vmatrix} = (6x+6)(6y-6)$$

at $(0,0)$

$$D(0,0) = \begin{vmatrix} 6 & 0 \\ 0 & -6 \end{vmatrix} = -36 < 0$$ so there is saddle point at $(0,0)$

at $(0,2)$

$D(0,2) = 6(12-6) = 36 > 0$ and $f_{xx}(0,2) = 6 > 0$

so there is a local minima at $(0,2)$

at $(-2,0)$

$D(-2,0) = (-12+6)(-6) = 36 > 0$

and $f_{xx}(-2,0) = 6 \times -2 + 6 = -6 < 0$

so there is a local maxima at $(-2,0)$

at $(-2,2)$

$D(-2,-2) = (-12+6)(-12-6) = -36 < 0$

so there is another saddle point at $(-2,2)$

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(b) $f(x,y) = 9x^3 + \frac{y^3}{3} - 4xy$

$f_x = 27x^2 + 0 - 4y$ $\quad f_y = \frac{3y^2}{3} - 4x$ $\quad f_{yy} = 2y$

$f_{xx} = 54x$ $\quad f_{yx} = -4$

$$D(x,y) = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = \begin{vmatrix} 54x & -4 \\ -4 & 2y \end{vmatrix}$$

for critical points : $f_x = 0 \Rightarrow 27x^2 = 4y$

$f_y = 0 \Rightarrow y^2 = 4x \Rightarrow x = \frac{y^2}{4}$

$27\left(\frac{y^2}{4}\right)^2 = 4y$

$27y^4 - 16y = 0$

$y(27y^3 - 64) = 0 \Rightarrow y = 0 \quad y^3 = \frac{64}{27}$

$y = \frac{4}{3}$

$y = 0 \Rightarrow x = 0$

$y = \frac{4}{3} \Rightarrow x = \frac{16/9}{4}$

$x = 4/9$

point is $(0,0)$ $(\frac{4}{9}, \frac{4}{3})$

at $(0,0)$ $\quad D(0,0) = \begin{vmatrix} 0 & -4 \\ -4 & 0 \end{vmatrix} = 0 - (16) = -16 < 0$

$\Rightarrow$ there is a saddle point at $(0,0)$

at $(4/9, 4/3)$ $\quad D(\frac{4}{9}, \frac{4}{3}) = \begin{vmatrix} 54 \times \frac{4}{9} & -4 \\ -4 & 2 \times \frac{4}{3} \end{vmatrix}$

$= \begin{vmatrix} \frac{18 \times 4}{3} & -4 \\ -4 & \frac{8}{3} \end{vmatrix} = \frac{576}{9} - 16 > 0$

$f_{xx}(4/9, 4/3) = 54 \times 4/9 > 0$

So there is a local minima at $(4/9, 4/3)$.

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(c) $f(x,y) = x^2+y^2-2x-6y+10$

$f_x = 2x-2 \qquad f_y = 2y-6 \qquad f_{yx} = 0$

$f_{xx} = 2 \qquad f_{yy} = 2$

$$D(x,y) = \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} = 4$$

<u>for critical points</u> $f_x = 0 \Rightarrow x = 1$ $\qquad (1,3)$

$f_y = 0 \Rightarrow y = 3$

<u>at (1,3)</u> $D(1,3) = 4 > 0$

$f_{xx} = 2 > 0$

$\Rightarrow$ there is local minima at (1,3).

(d) $f(x,y) = x^4 - 4xy + y^4$

$f_x = 4x^3 - 4y \qquad f_y = -4x + 4y^3 \qquad f_{yy} = 12y^2$

$f_{xx} = 12x^2 \qquad f_{yx} = -4$

$$D(x,y) = \begin{vmatrix} 12x^2 & -4 \\ -4 & 12y^2 \end{vmatrix}$$

<u>for critical points</u> $f_x = 0 \Rightarrow 4x^3 = 4y \Rightarrow y = x^3$

$f_y = 0 \Rightarrow 4x = 4y^3 \Rightarrow x = y^3$

solving these two relations we get the points

$(0,0)$ $(1,1)$ $(-1,-1)$

<u>at (0,0)</u> $D(0,0) = -16 < 0 \Rightarrow (0,0)$ is saddle point

<u>at (1,1)</u> $D(1,1) = 128 > 0$ and $f_{xx} = 12 > 0$

$\Rightarrow$ local minima at (1,1)

<u>at (-1,-1)</u> $D(-1,-1) = 128 > 0$ and $f_{xx} = 12 > 0$

$\Rightarrow$ local minima at (-1,-1)

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4 (a) $f(x,y) = x^2y + xy + y$

point is $(3,2)$, $u = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}} i + \frac{1}{\sqrt{2}} j$

$\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = 1$

which is unit vector

$$\left.\begin{aligned} f_x &= 2xy + y \\ f_y &= x^2 + x + 1 \end{aligned}\right\} \Rightarrow \begin{aligned} f_x(3,2) &= 2 \cdot 3 \cdot 2 + 2 \\ &= 14 \\ f_y(3,2) &= 9 + 3 + 1 \\ &= 13 \end{aligned}$$

$$D_{\hat{u}} f(3,2) = f_x(3,2) \cdot \frac{1}{\sqrt{2}} + f_y(3,2) \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{14}{\sqrt{2}} + \frac{13}{\sqrt{2}}$$

$$= \frac{27}{\sqrt{2}}$$

(b) $f(x,y) = x^2y + xy + y$

point is $(3,2)$ $\quad u = (1,2) = 1i + 2j$

which not unit

$\|u\| = \sqrt{1^2 + 2^2} = \sqrt{5}$

so $\hat{u} = \frac{1}{\sqrt{5}} i + \frac{2}{\sqrt{5}} j$ is unit vector

$$D_{\hat{u}}(3,2) = f_x(3,2) \frac{1}{\sqrt{5}} + f_y(3,2) \frac{2}{\sqrt{5}}$$

$$= \frac{14}{\sqrt{5}} + \frac{13 \times 2}{\sqrt{5}} = \frac{40}{\sqrt{5}}.$$

(c) $f(x,y) = x^2y + xy + y$

point is $(3,2)$, $u = (2,1) = 2i + 1j$ which not unit

$\|u\| = \sqrt{2^2 + 1^2} = \sqrt{5}$

$\hat{u} = \frac{2}{\sqrt{5}} i + \frac{1}{\sqrt{5}} j$ is unit vector

$$D_{\hat{u}}(3,2) = f_x(3,2) \frac{2}{\sqrt{5}} + f_y(3,2) \frac{1}{\sqrt{5}}$$

$$= \frac{14 \times 2}{\sqrt{5}} + \frac{13 \times 1}{\sqrt{5}} = \frac{28 + 13}{\sqrt{5}} = \frac{41}{\sqrt{5}}$$

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5 $f(x,y) = 5 - 2x^2 - \frac{1}{2}y^2$

point is $(3,4)$

$u = \cos\frac{\pi}{4}\, i + \sin\frac{\pi}{4}\, j$

$u = \frac{1}{\sqrt{2}} i + \frac{1}{\sqrt{2}} j$ which is unit vector $\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = 1$

$f_x = -4x$ $\qquad f_x(3,4) = -4 \times 3 = -12$

$f_y = -\frac{2}{2}y = -y$ $\qquad f_y(3,4) = -4$

$D_u(3,4) = f_x(3,4)\frac{1}{\sqrt{2}} + f_y(3,4)\frac{1}{\sqrt{2}}$

$= -\frac{12}{\sqrt{2}} - \frac{4}{\sqrt{2}} = \frac{-16}{\sqrt{2}} = \frac{-8 \times \sqrt{2} \cdot \sqrt{2}}{\sqrt{2}} = -8\sqrt{2}$ Ans

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